

Comparative Cytoarchitectural Analyses of Striate and Extrastriate Areas in Hominoids

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The visual cortex is the largest sensory modality representation in the neocortex of humans and closely related species, and its size and organization has a central role in discussions of brain evolution. Yet little is known about the organization of visual brain structures in the species closest to humans—the apes—thus, making it difficult to evaluate hypotheses about recent evolutionary changes. The primate visual cortex is comprised of numerous cytoarchitecturally distinct areas, each of which has a specific role in the processing of visual stimuli. We examined the histological organization of striate (V1) and 2 extrastriate (V2 and ventral posterior) cortical areas in humans, 5 ape species, and a macaque. The cytoarchitectural patterns of visual areas were compared across species using quantitative descriptions of cell volume densities and laminar patterns. We also investigated potential scaling relationships between cell volume density and several brain, body, and visual system variables. The results suggest that interspecific variability in the cytoarchitectural organization of visual system structures can arise independently of global brain and body size scaling relationships. In particular, species-specific differences in cell volume density seem to be most closely linked to the size of structures in the visual system.

Keywords: allometry, evolution, gray level index, hominoids, visual cortex

Introduction

Cortical areas devoted entirely or mainly to vision comprise 23% of the adult modern human neocortex (Van Essen and Drury 1997) and 55% of the macaque neocortex (Felleman and Van Essen 1991). Physiological and histological investigations have lead to the identification of over 25 visual areas in macaques (Felleman and Van Essen 1991; Hof and Morrison 1995; Van Essen 2004). The catarrhines—apes, humans, and Old World monkeys—possess a particularly specialized visual system. Like other primates, they possess well-developed stereoscopic vision, but catarrhines are further characterized by routine trichromatic color vision (Jacobs and Kaas 2007; Ross and Martin 2007). Recently, functional magnetic resonance imaging has made it possible to map and identify visual areas in the human cerebral cortex, and in some cases, candidates for homologous areas between the human and monkey brain have been identified (DeYoe et al. 1996).

Although the size and organization of the human visual cortex has played a fundamental role in discussions of brain evolution (e.g., Dart 1925; Holloway 1972; Jerison 1975; Falk 1980; Holloway et al. 2003), little is known about the detailed anatomy and physiology of the visual areas in the species phylogenetically closest to humans—the apes. Only rarely have there been opportunities to conduct physiological studies of ape visual cortex (e.g., Tigges and Tigges 1979; Vital-Durand and Blakemore 1981). Nonetheless, comparative anatomical investigations could provide insight into whether the human visual system displays any evidence of evolutionary specialization (Preuss 2004). The focus of the current study is to improve the representation of apes in cytoarchitectural data sets, which, in combination with independent molecular evidence about their relationships, will make it possible to generate hypotheses about the evolution of the higher primate visual cortex.

In addition, this study explores the possibility of scaling relationships between neuronal volume density (indicated by the gray level index, GLI) and several brain, body, and visual system variables. GLI is highly correlated with neuronal volume density (Wree et al. 1982) a component of which is neuronal numerical density (additional information about the relationship of GLI values to neuronal density can be found in the Supplementary Material). Several studies have demonstrated that, across a range of mammalian species, there is a negative allometric relationship between brain size and neuron density in the neocortex, which follows a $-1/3$ power law (Nissl 1898; Tower 1954; Haug 1987; Prothero 1997). In the current study, the possibility that visual area GLI values vary according to the size of functionally related visual system structures is explored.

Materials and Methods

Specimens and Tissue Preparation

The sample for comparative cytoarchitectural analysis comprised a total of 9 brains representing 7 catarrhine species, that is, one each of *Macaca fascicularis* (long-tailed macaque), *Hylobates lar* (lar gibbon), *Pongo pygmaeus* (orangutan), *Gorilla gorilla* (gorilla), and *Pan paniscus* (bonobo), and 2 each of *Pan troglodytes* (chimpanzees) and *Homo sapiens* (humans).

Included were sections from the left hemispheres of adult specimens. The age and sex distribution is shown in Table 1. Specimens in the study belong to the Zilles comparative neuroanatomy collections at the C&O Vogt Institute of Brain Research in Düsseldorf, Germany. The

Table 1
Samples used in analyses of V1, V2, and VP

Species	code	archive number	sex	age (yrs)	body mass (kg)	brain mass (g)	EQ ^e	neocortex volume (cm ³)	left V1 vol (mm ³)	left LGN vol (mm ³)	optic nerve cross sectional area (mm ²) ^f	eye half surface area (mm ²) ^f
<i>Gorilla gorilla</i>	ggy	YN 82-140	F	20	85	376	1.20	254	4044	150	17.6	1774
<i>Hylobates lar</i>	hld	Disco	F	22	7	120	2.54	77	2292	90	12.7	1299
<i>Homo sapiens</i> ^{a,b}	hs5	54491	F	79	63	1350	5.41	974	7587	186	22.8	1855
<i>Homo sapiens</i> ^{a,b}	hs6	6895	F	79	63	1110	4.45	974	7013	156	22.8	1855
<i>Macaca fascicularis</i>	mf2	ma22	M	3	3	58	2.31		1357	46	8.4	985
<i>Pongo pygmaeus</i>	ouy	YN 85-38	M	16.5	58	369	1.56	269	3504	92	16.1	1282
<i>Pan paniscus</i> ^c	ppz	Zahlia	F	11	33	324	2.09	279	5687	130		
<i>Pan troglodytes</i>	ptb	Bathsheba	F	24	80	360	1.20	263	4705	168	16.0	1446
<i>Pan troglodytes</i> ^d	ptd	1548	NA	NA	51	387	1.82	198	2799	86	16.0	1446

^aUsed same sex species mean value for body weight (Zilles 1972).

^bUsed combined sex mean human neocortex value ($n = 8$) based on unpublished data provided by Carol MacLeod.

^cUsed same sex species mean value for body weight (Jungers and Susman 1984).

^dUsed combined sex species mean values for brain and body weight (Herdon et al. 1999.).

^eEncephalization quotient (EQ) after Martin (1981) and Ruff et al. (1997).

^fSpecies mean data from Stephan and Frahm 1981.

macaque brain was provided by Hans-Jürgen Bidmon, and is also at the C&O Vogt Institute.

The human and nonhuman hominoid brains from the Zilles collection were immersion fixed with either 4% formaldehyde or Bodian's solution (a mixture of formalin, glacial acetic acid, and ethanol), within a post mortem interval of less than 36 h after death for the human brains and less than 12 h after death for the nonhuman hominoid brains. Brain were embedded in paraffin and serially sectioned along the coronal plane at a thickness of 20 μ m. Sections used in this analysis were stained for cell bodies using silver impregnation according to the technique described by Merker (1983), which is based on the procedure of Gallyas (1971). Merker stain is ideal for quantitative cytoarchitectural analysis due to high staining intensity and contrast. The macaque brain was perfusion fixed with 4% formaldehyde in phosphate buffer, embedded in paraffin and serially sectioned along a coronal plane at 20 μ m, and Merker stained for cell bodies.

Grey Level Index Acquisition

GLI values were obtained to quantify the laminar organization of cortical areas V1, V2, and ventral posterior (VP), which were identified using cytoarchitectural criteria. These 3 early visual areas were chosen for their contiguity with each other and their recognizability. Additional information about the identification and sampling of cortical areas V1, V2, and VP is described in the Supplementary Material. The GLI encompasses the volume density of cell bodies and their sizes per unit volume of cerebral cortex. GLI values measure the proportion of tissue volume that is neuronal cell nuclei, glial cell nuclei, and endothelial cell nuclei versus the proportion of tissue volume that is occupied by neuropil. As the volume fraction of glial and endothelial cells is small and does not vary significantly across cortical layers, the GLI is an estimate of neuronal volume density (Wree et al. 1982). Differential shrinkage of cell bodies versus neuropil may occur, although it is expected to be lower than 9% (Kretschmann et al. 1982). The steps involved in obtaining GLI values for a region of interest (ROI) are described here.

First, high-resolution images of cell body stained sections were obtained. For each specimen, per cortical area, 5 adjacent sections equally spaced 300–400 μ m apart were quantified. A rectangular image of the ROI was obtained by an automatic scanning technique using an image analysis system (KS 400; Zeiss, Oberkochen, Germany). The ROI is viewed one image frame at a time through a light microscope (Zeiss Planapo) at 10×1.25 , and digitized using a digital camera (AxioCam MRm, Zeiss, Germany). A motorized stage controlled by KS400 software is used to move between image frames. Thus, the ROI is actually a mosaic of several adjacent rectangular image frames, each an 8-bit gray-scale image 713 $\times$ 537 μ m in size and 1376 $\times$ 1036 pixels in spatial resolution (i.e., each square pixel was 0.518 μ m in length).

Note that the GLI values reported here are not directly comparable to previous reports of cortical area GLI values (e.g., Semendeferi et al. 1998, 2001; Sherwood et al. 2004; Amunts et al. 2007) due to differences in staining, magnification, and/or segmentation of cell bodies. Although a magnification of 4×1.26 is routine and sufficient for producing GLI images in most cases, it was found that the densest layers of nonhuman primate brains—for example, V2 granular layer IV, perhaps the densest layer in the cerebral cortex (von Economo 1929)—required a higher magnification in order to ensure accurate segmentation of neurons during image processing. Thus, the procedure for segmenting cell bodies from neuropil was adjusted so as to be optimized for this higher magnification. Although a smaller field of view results in lower, more precise, and less biased GLI values (Wree et al. 1982), the effect of this adjustment overcompensates for this effect through the identification of smaller particles than in previous studies; therefore, the human GLI values reported here are about twice as high as those reported by Amunts et al. (2007).

Next, each image frame was converted to binary values using adaptive thresholding, and further subdivided into a grid of 81×61 measuring fields (17×17 pixels per square, comprised of 16 pixels which are measuring field and 1 pixel which is border) (Wree et al. 1982; Schleicher and Zilles 1990). This sampling procedure resulted in measuring field size of $8.3 \times 8.3 \mu$ m. For each measuring field, a GLI value was obtained, and these were combined in a composite data matrix called the GLI image.

GLI profiles were extracted from a GLI images as follows. On a GLI image, 2 contour lines were manually traced at the borders between cortical layers I and II, and between cortical layer VI and the white matter (Fig. 1). GLI profiles represent variation in cell volume density across cortical layers. Profiles were extracted using a MatLab v. 7.1 (2005, The MathWorks, Inc., Natick, MA) based routine, by calculating equally spaced profile lines (traverses) perpendicular to the 2 contours and parallel to the cortical columns (Jones et al. 2000). Traverses were calculated to be curved, to best represent the actual shape of cortical columns. Cortical thickness is not constant, so each profile is standardized to encompass 101 values corresponding to cortical depth, ranging from 0% (border between layer I and layer II) to 100% (border between layer VI and white matter), which is achieved by resampling for linear interpolation.

GLI profiles were quantitatively represented by a set of 10 feature vectors describing the shape of the curve, based on its central moments (Amunts et al. 1997). These data formed the basis of all the subsequent multivariate analyses.

Layerwise GLI Values

Within each cortical area, laminar borders were determined using cytoarchitectonic criteria (additional information about the laminar boundaries is described in the Supplementary Material), and the mean

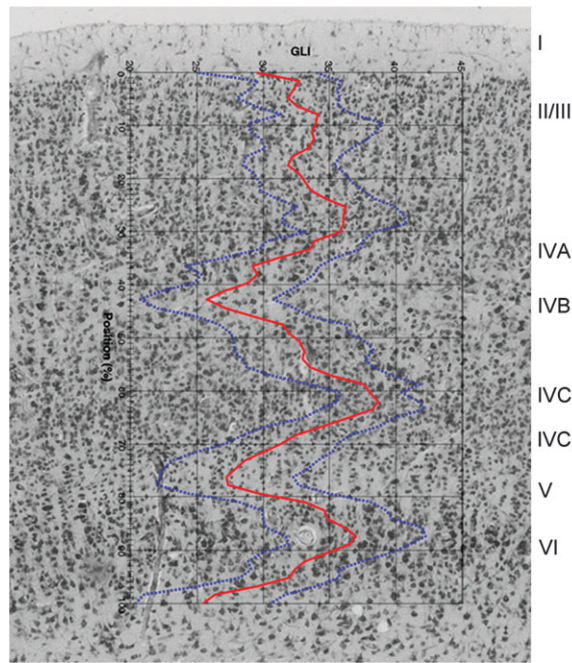


Figure 1. Procedure for obtaining layerwise GLI values. GLI mean profile graphs were extracted and superimposed over histological images to determine the relative depth of the layer divisions.

GLI value for each layer was calculated by subdividing the mean GLI profiles into segments corresponding to cortical layers and sublayers (except layer I). For each ROI, the maxima and minima of the mean GLI profile were matched to cytoarchitecturally identified layer divisions (Fig. 1C). This was achieved by fitting a translucent mean profile graph over 10×1.25 magnification histological image of representative portions of cortex, and the breakpoints between lamina were recorded.

Comparative Statistical Methods

Bivariate and multivariate statistical methods were used to examine possible scaling relationships between V1, V2, and VP mean GLI and 2 sets of macroanatomical variables: brain and body size variables, and visual system variables (Table 1). First correlations between V1, V2, and VP mean GLI and the brain, body, and visual system variables were examined, and these were then followed by regression analyses to determine the slope of the scaling relationships.

The selection of the 4 brain and body size variables was based on Sherwood et al. (2004) and included brain mass (BRAIN), body mass (BODY), encephalization quotient (EQ), and neocortex volume (NEOCORTEX). Because it is possible that scaling relationships could occur at the level of sensory modality, correlations were also conducted to include 4 visual system variables: left lateral geniculate nucleus (LGN volume) (LGN), left V1 volume (V1 VOL), half surface of eye area (EYE), and optic nerve cross sectional area (OPTIC NERVE).

In most cases, these data were available for the individual specimens (de Sousa et al., unpublished data), but where not available, species means were used from the literature (Stephan et al. 1981). EQs were calculated using Ruff et al.'s (1997) formula, which is based on Martin (1981):

$$EQ = \text{brain mass} / \left(11.22 \times \text{body mass}^{3/4} \right)$$

Prior to statistical analyses, each of the brain, body, and visual system variables was transformed to approach the linear dimensions of the GLI data (Sherwood et al. 2005). GLI was compared with the cubic root of the volume measurements V1, LGN, NEO, and BRAIN, and the mass measurement BODY (a proxy for body volume). GLI was compared with the square root of the area measurements OPTIC NERVE and EYE. EQ was not transformed because it is a residual value. Next, the GLI

values and the brain, body, and visual system variables were $\log_{10}$ transformed to facilitate the use of statistical methods to calculate a best-fit linear relationship.

Simultaneous correlations were done for the 3 cortical areas, V1, V2, and VP, tripling the probability of false positives, so in order to obtain significance at alpha level 0.05 a Bonferroni correction was applied and only P values of $0.05/3 = 0.0167$ were considered to be statistically significant.

Mean GLI Values

First, nonparametric correlation analyses (Spearman's rho) were performed to investigate the relationships between mean cortical area GLI and the sets of brain and body size and visual system variables. After correlations were determined, each brain, body, and visual system variable that was significantly correlated with the mean GLI for V1, V2, or VP was treated as the independent variable in regressions in which mean GLI was the dependent variable. Slopes were calculated in SMATR (Warton et al. 2006) from reduced major axis (RMA) regressions, a method which assumes that the error variance of the X and Y variates is equal.

Because the variables had been put into the same dimension and then log transformed, the slope of the linear regressions corresponds to scaling coefficients. Therefore, an isometric relationship would be represented by slope of 1, a positive allometric scaling relationship would be represented by a slope of greater than 1, and a negative allometric scaling relationship would be represented by a slope of less than 1.

Mean GLI Profiles

Multivariate methods were used to analyze mean cortical GLI profiles. Mean cortical profiles were analyzed for each specimen rather than averaged between members of the same species because including multiple specimens tends to blur profile data (Schleicher et al. 2000). Euclidean distances were calculated to summarize cytoarchitectural distances within and between species, per cortical area. Euclidean distances were also used to compare distances within cortical areas to distances between cortical areas for the entire group of specimens. Principal components analysis (PCA) was used to explore the contribution of different GLI feature vectors to phylogenetic differences in profile shape. Finally, discriminant function analysis was used to determine whether GLI profile data could be used to assign a pattern to each cortical area, across the range of species.

Results

Using the cytoarchitectural criteria described above, it was possible to locate areas V1, V2, and VP in each of the species studied (Figs 2 through 4).

Relative Laminar Widths

The species mean relative cortical layer widths are shown for areas V1, V2, and VP in Figure 5. In general, cortical layer widths were found to be quite similar across species, in all the cortical areas. The variation that exists in layer width patterning does not seem to reflect taxonomic relationships. This is in agreement with a previous study (Zilles and Rehkämper 1988) which did not identify much variation in V1 layer widths in a smaller hominoid sample.

Grey Level Index

Species mean cortical GLI values (Fig. 6) and layerwise GLI values (Fig. 7A, B, and C) are shown for areas V1, V2, and VP. For all visual areas studied, species mean GLI values generally decrease with phylogenetic proximity to *Homo sapiens*, with some notable exceptions. In V1, V2, and VP, humans have the lowest mean GLI value, which is consistent with the finding

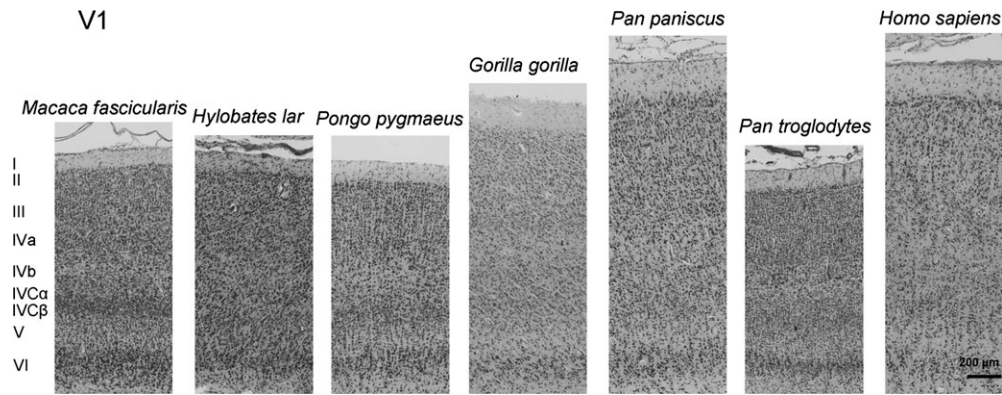


Figure 2. Cytoarchitecture of cortical area V1 in a macaque, a gibbon, an orangutan, a gorilla, a bonobo, a chimpanzee, and a human.

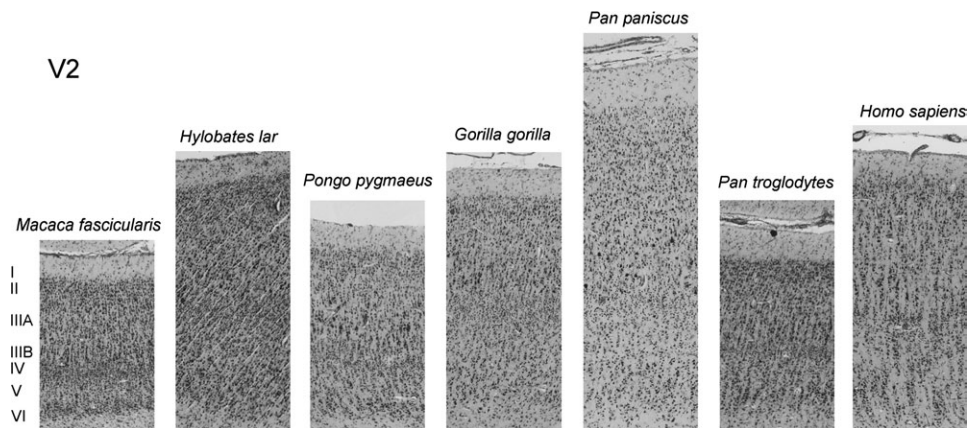


Figure 3. Cytoarchitecture of cortical area V2 in a macaque, a gibbon, an orangutan, a gorilla, a bonobo, a chimpanzee, and a human.

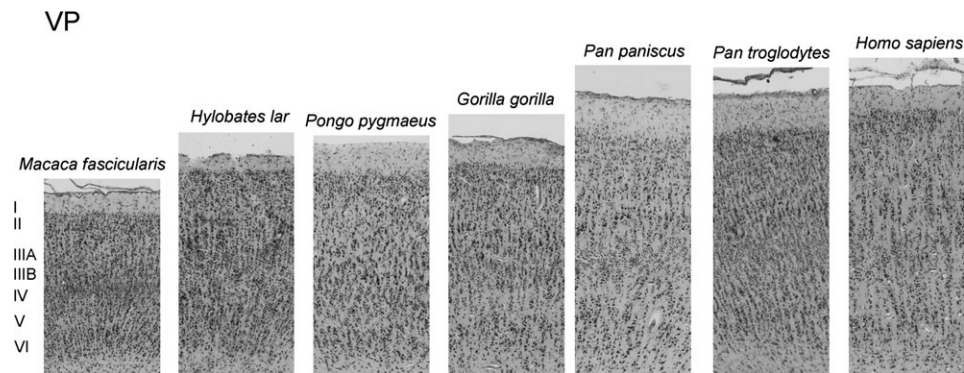


Figure 4. Cytoarchitecture of cortical area VP in a macaque, a gibbon, an orangutan, a gorilla, a bonobo, a chimpanzee, and a human.

that humans have the lowest neuronal density in striate and extrastriate cortical areas among anthropoid primates (Sherwood et al. 2007). In V1, *Macaca fascicularis* had the highest mean GLI values, although in V2 and VP, the mean GLI values are higher in *Hylobates lar*. *Pan troglodytes* GLI values diverge from the phylogenetic trend in that they are consistently high, in areas V1, V2, and VP, and in both individuals. The high GLI values of *Pan troglodytes* contrast with the much lower values of *Pan paniscus*.

Correlation analysis was performed on mean V1, V2, and VP GLI values and several brain and body and visual structure size

variables. Mean V1 GLI values were found to be negatively correlated with 3 visual system structures: OPTIC NERVE ($r = -1$, $P < 0.0001$), LGN ($r = -0.93$, $P = 0.0025$), and V1 VOL ($r = -0.96$, $P = 0.0004$), and with the volume of the brain structure NEOCORTEX ($r = -0.94$, $P = 0.0048$). Mean V2 GLI values were negatively correlated with OPTIC NERVE ($r = -0.94$, $P = .0048$) and V1 VOL ($r = -0.89$, $P = 0.0069$). Mean VP GLI values were also negatively correlated with OPTIC NERVE ($r = -0.94$, $P = 0.0048$) and V1 VOL ($r = -0.89$, $P = 0.0068$). No other correlations were found to be statistically significant.

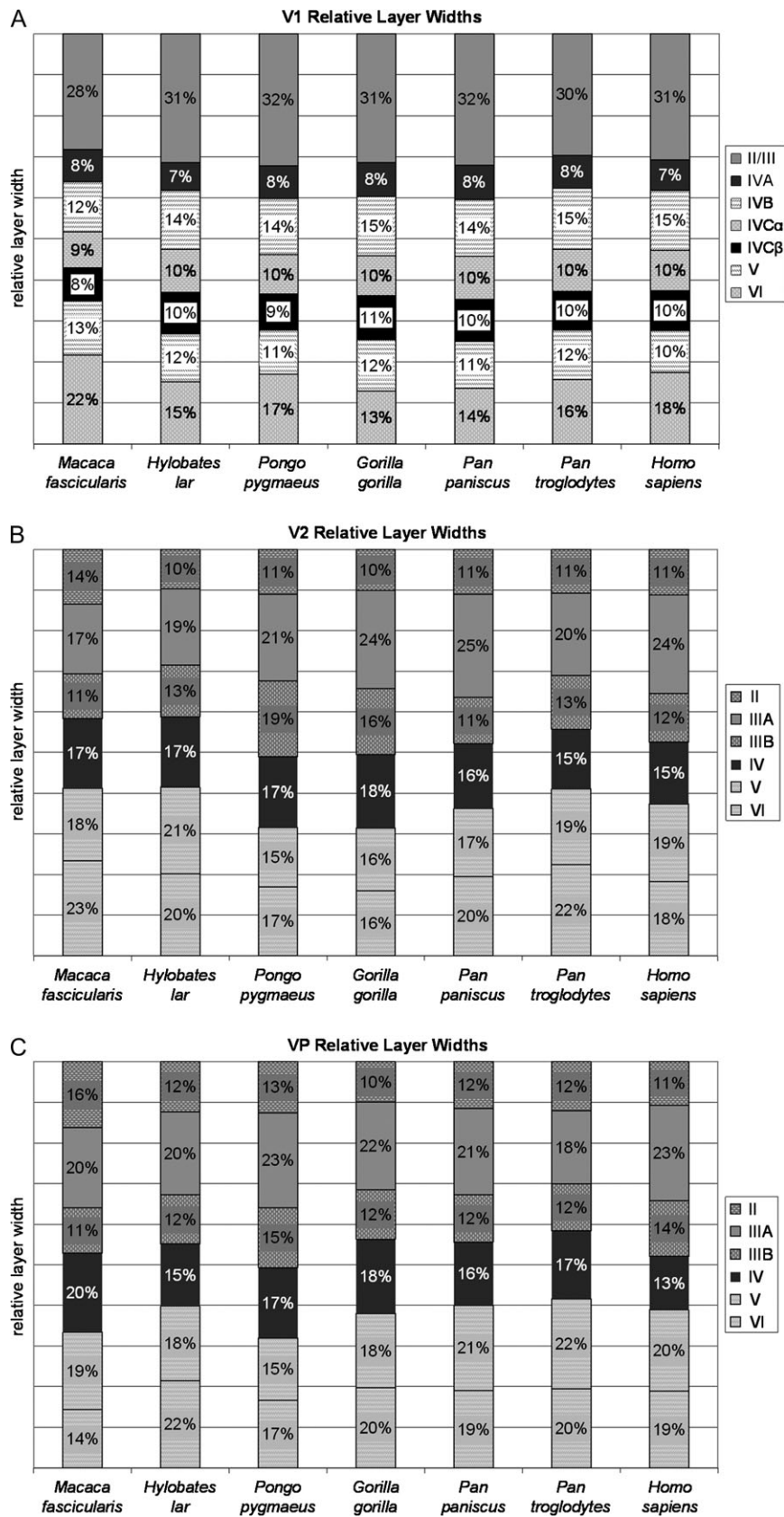


Figure 5. Species mean relative layer widths for areas V1 (A), V2 (B), and VP (C). Each bar represents the proportion of the total cortical thickness occupied by each layer.

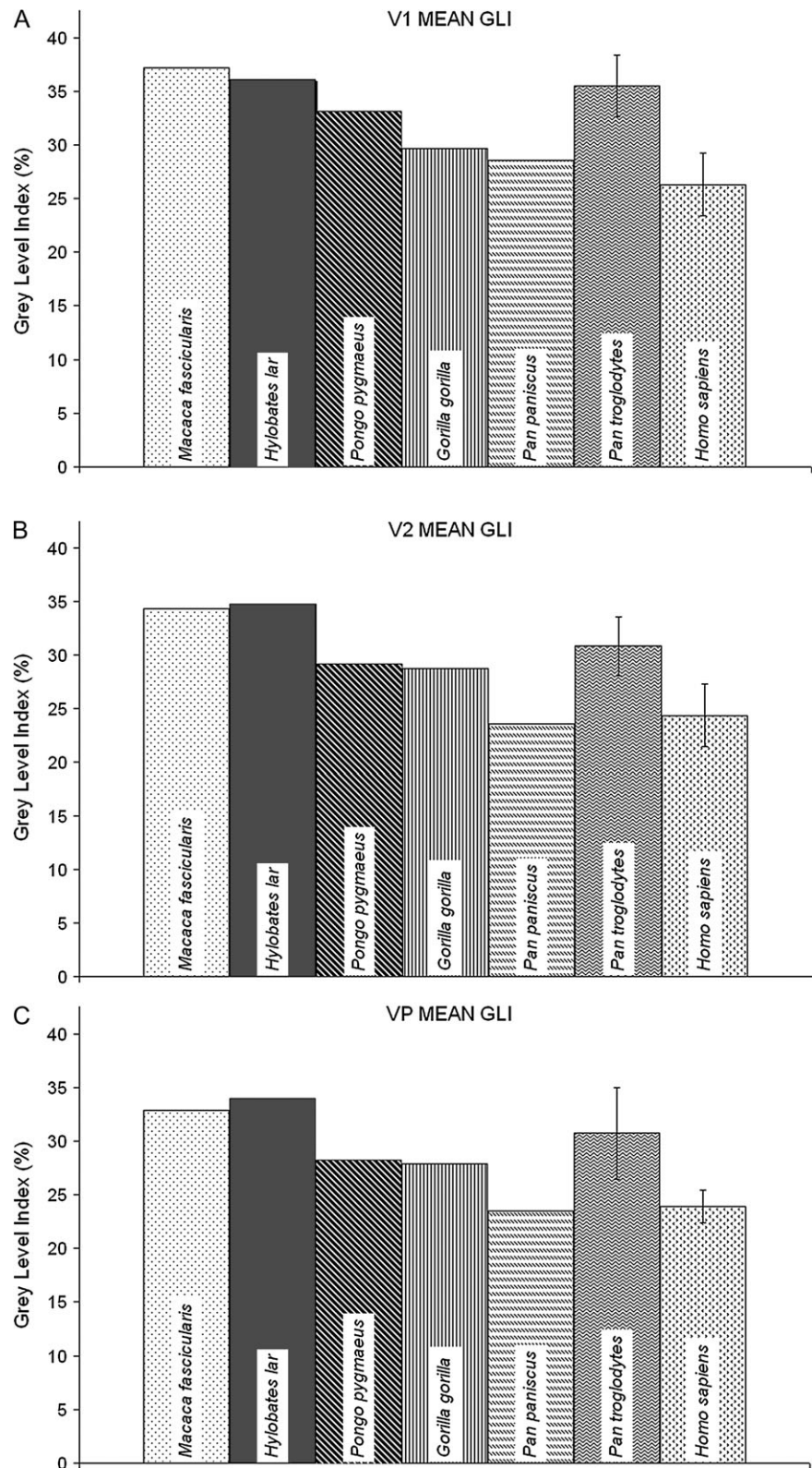


Figure 6. Species mean GLI values for areas V1 (A), V2 (B), and VP (C). Vertical bars denote 95% confidence intervals.

These correlations were followed by regression analyses to determine the slopes of the scaling relationships (Table 2 and Fig. 8). All scaling relationships were found to

be negatively allometric (i.e., with a slope of less than 1). This means that GLI decreases at a slower rate than changes in the size of the macrostructures. Thus, for example, as

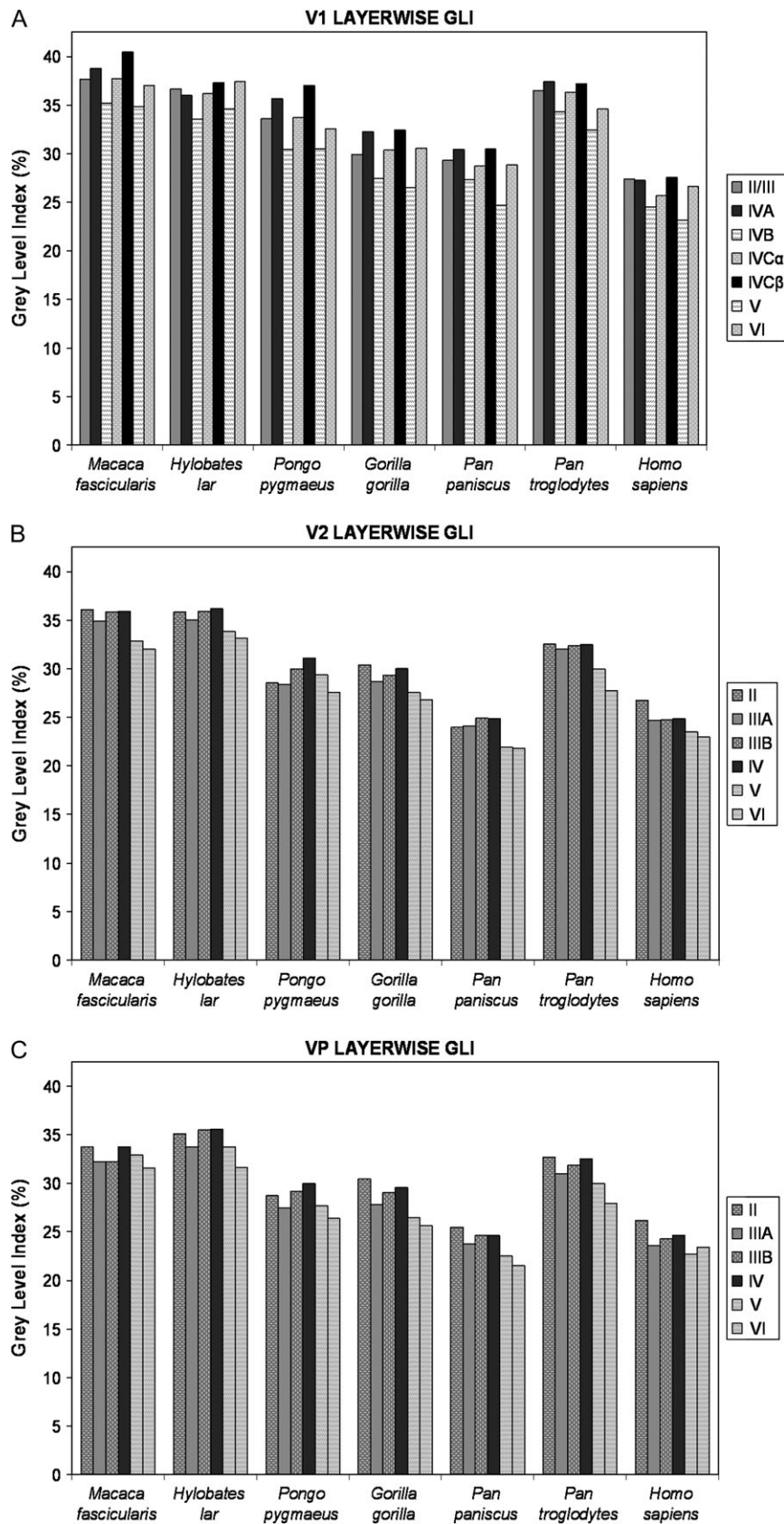


Figure 7. Species mean layerwise GLI values for areas V1 (A), V2 (B), and VP (C).

Table 2

RMA regressions of V1, V2, and VP GLI values on brain and visual system variables.

Y	X	n	R ²	p	Slope	Lower 95% CI	Upper 95% CI	Intercept	Lower 95% CI	Upper 95% CI
GLI V1	NEOCORTEX VOL	6	0.677	0.044	-0.47	-0.98	-0.23	1.88	1.57	2.18
	V1 VOL	7	0.788	0.008	-0.24	-0.39	-0.14	2.35	1.90	2.79
	LGN VOL	7	0.610	0.038	-0.90	-1.76	-0.46	2.12	1.68	2.56
	OPTIC NERVE	6	0.704	0.037	-0.79	-1.58	-0.39	1.98	1.63	2.33
GLI V2	V1 VOL	7	0.821	0.005	-0.27	-0.43	-0.17	2.43	1.96	2.90
	OPTIC NERVE	6	0.777	0.020	-0.77	-1.43	-0.42	1.93	1.63	2.23
GLI VP	V1 VOL	7	0.783	0.008	-0.26	-0.43	-0.15	2.37	1.88	2.86
	OPTIC NERVE	6	0.707	0.036	-0.75	-1.51	-0.38	1.91	1.58	2.25

Notes: Regression were calculated for variables found to be correlated to GLI. These were converted into the same dimension as the GLI data, and then the data were all log transformed.

V1 volume increases, V1 GLI decreases, but at a slower rate.

GLI Profile Analysis

Mean GLI profiles were calculated for each specimen to represent the laminar patterns of areas V1, V2, and VP (Figs 9–11). GLI profiles were characterized quantitatively by conversion into 10 feature vectors based on the central moments of the mean GLI curve: the mean GLI (mean_{y.o}), the mean cortical depth (mean_{x.o}), the standard deviation (sd_{y.o}), the skewness (skew_{y.o}), the kurtosis (kurt_{y.o}), and the same parameters for the first derivative of each mean profile (mean_{y.d}, mean_{x.d}, sd_{y.d}, skew_{y.d}, kurt_{y.d}); for more details see Amunts et al. (2003). These data were then converted to Z-transforms in the subsequent statistical analyses so that the variables would each have equal weights.

Multidimensional scaling was performed to compute distances from 4 multivariate data sets of the 10 feature vectors as a data reduction technique. Identical procedures were followed for the following data sets: V1, V2, VP, and all areas combined. For each of the areas V1, V2, and VP, the mean Euclidean distances between species exceeded the distance within species (Fig. 12). The Euclidean distance model plot (Fig. 13) demonstrates the extent to which Euclidean distance data can be used to differentiate groups. In some cases, clusters differentiate cortical areas; in other cases, they differentiate species. For example, an area V1 cluster is differentiated mainly on the basis of Dimension 1 (*x* axis). Also, human extrastriate areas form a cluster in the negative Dimension 1, negative Dimension 2 quadrant of the graph.

Because this indicates that in some cases interspecific (or interindividual) differences eclipse differences in cortical pattern (Table 3A), Euclidean distances between areas per specimen were averaged to determine the mean distances between different cortical areas (Table 3B). Area V1, which has the most distinct laminar pattern, had the greatest mean distances from the other 3 areas. Interestingly, Euclidean distances correspond to topographical distances along the visual processing hierarchy. V1 is closest to V2, and farthest from VP. The smallest mean distance between areas was between the adjacent extrastriate areas V2 and VP.

A stepwise discriminant analysis was performed to determine whether assignment to cortical areas V1, V2, and VP could be predicted on the basis of GLI profile data alone. All 10 feature vectors were entered (with block entry of variables), but only 3 of these variables were included in the reduced model: skew_{y.d}, sd_{y.d}, and kurt_{y.o}. The classification resulted in 100% accurate predicted group membership to V1. The results for extrastriate

areas were less accurate. Those assigned to V2 included 66.7% of V2, plus 22.2% of VP. Those predicted to be VP included 77.8% of VP, plus 33.3% of V2. Although GLI feature vector data are not sufficient for assigning cortical areas across the range of taxa, it is interesting to note that cortical area classification into striate versus extrastriate cortex can be accomplished across taxonomic groups on the basis of only 3 features.

A PCA was conducted to summarize the variance and to explore the contributions of the different feature vectors to differentiating cortical areas. The first 3 components had eigenvalues above 1, and combined explained 83.3% of the variance. The contributions of each of the feature vectors to the first 3 components are shown in Table 4. Although V1 values cluster on Factor 1, none of the first 3 factors was alone able to completely separate any of the cortical areas. For factor 1, the greatest positive loading was mean_{x.d}, followed by mean_{x.o}, mean_{y.o}, sd_{y.d}, and sd_{y.o}. The greatest negative loading was skew_{y.d}, followed by kurt_{y.o}, mean_{y.d}, skew_{y.o}, and kurt_{y.d}. A plot of the strongest positive and negative loadings separates V1 from the extrastriate areas, although V2 and VP values form 2 overlapping clusters (Fig. 14).

Discussion

Correlations and Scaling Relationships

Overall, striate and extrastriate cortex GLI values scale to visual system variables, particularly V1 volume and optic nerve cross sectional area. Differently, the visual cortex GLI values do not significantly correlate with brain and body variables. The only statistically significant correlation between any of the brain or body size variables and mean GLI is the correlation of V1 GLI to neocortex size (but note that much of the neocortex is visual in function). In contrast, there were 7 instances of significant correlations between visual area GLI values and visual system macrostructure sizes. Therefore, as discussed below, there may be a general pattern of scaling between cytoarchitectonic and gross-level anatomical variables within the visual system.

At first glance, the findings of this study seem surprising given the more general relationship between neuron numerical density and brain size. Across a range of mammalian species, there is a negative allometric relationship between brain size and neuron numerical density in the neocortex, which follows a $-1/3$ power law (Tower 1954; Haug 1987; Prothero 1997). This scaling relationship has been explained functionally and theoretically as a constituent of a model which describes that, regardless of brain size, cortical areas maintain a similar amount of influence over one another (Changizi 2001).

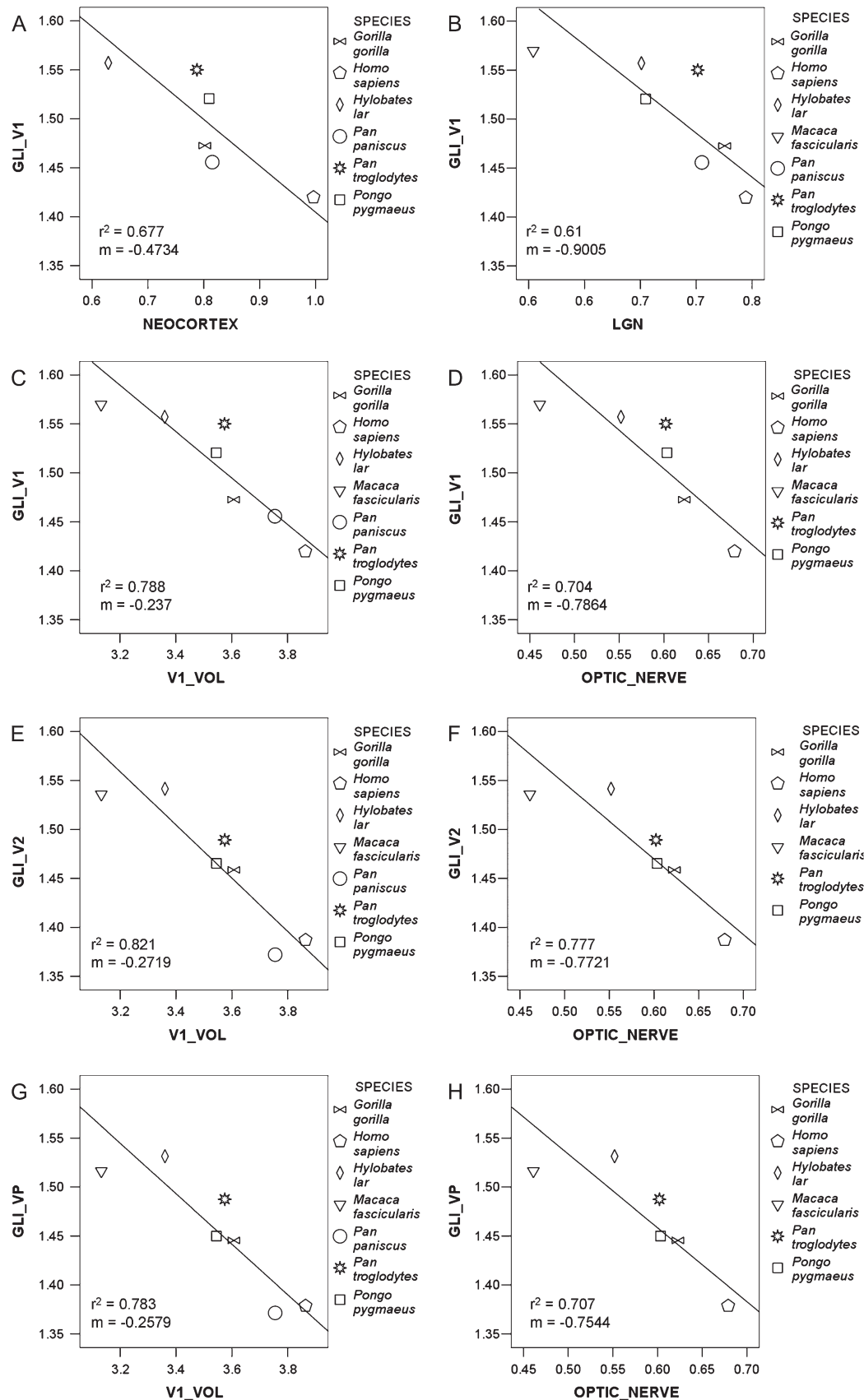


Figure 8. RMA regressions of species mean GLI values on brain and body size variables. V1 GLI as a function of neocortex volume (A), LGN volume (B), V1 volume (C), and optic nerve volume (D); V2 GLI values as a function of V1 volume (E); and optic nerve volume (F); and VP GLI values as a function of V1 volume (G); and optic nerve volume (H).

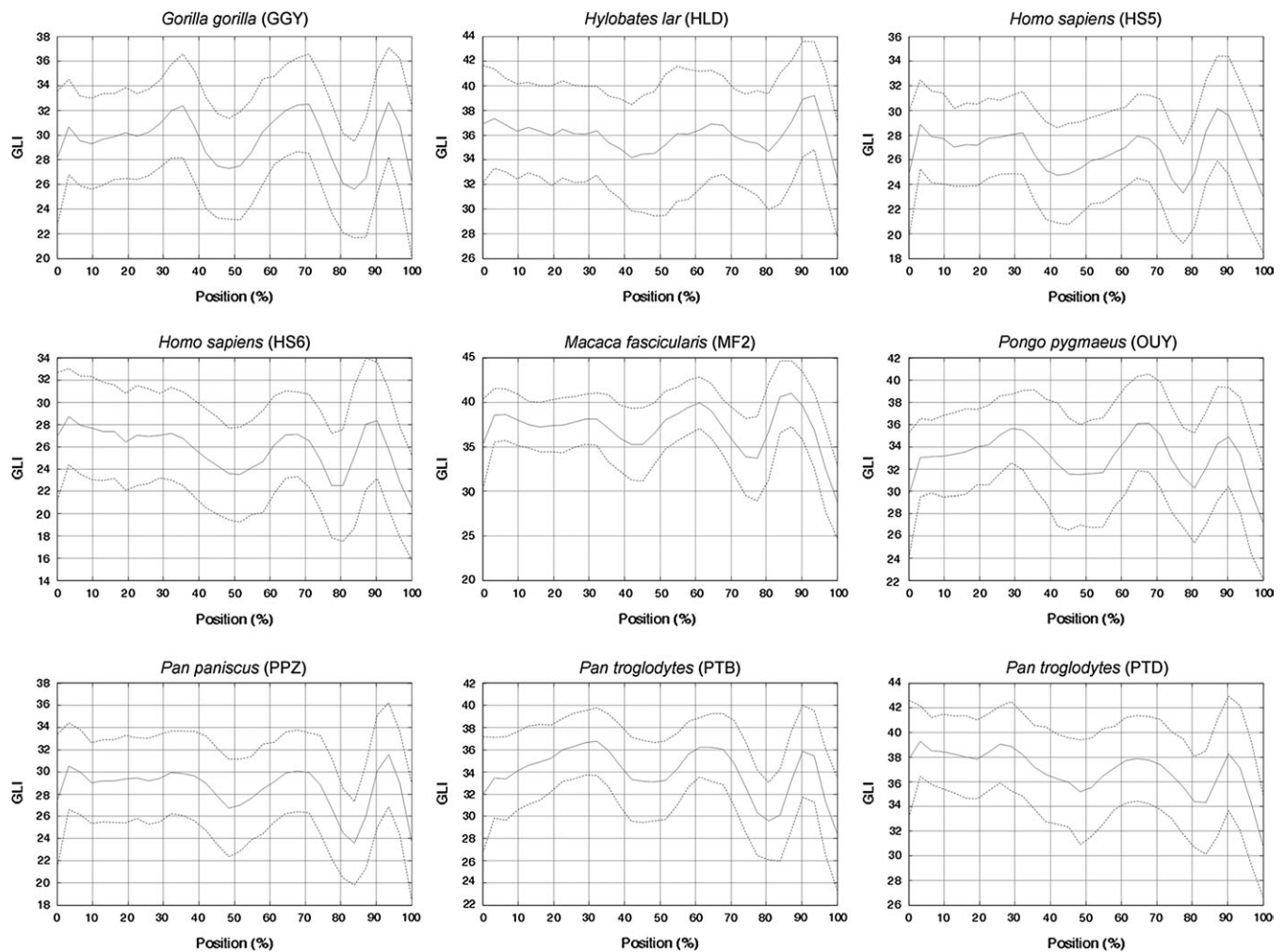


Figure 9. Mean V1 GLI profiles of individual specimens (solid lines) with standard deviation indicated (dotted lines).

However, the finding that cortical area GLI values do not correlate significantly with any brain or body size variables is actually congruent with previous studies. In hominoids, neither GLI nor neuron numerical density of areas 10, 13, and 4 scale to brain weight, although it is possible that the hominoid sample is not large enough to exhibit such scaling relationships (Sherwood and Hof 2007).

Interestingly, scaling relationships seem to differ between cortical areas. In primates, the neuron numerical density of V1 is about double that of other cortical areas, resulting in a grade-level shift in V1 not seen in the smooth scaling trends for other cortical areas (Cragg 1967; Rockel et al. 1980). For a single anthropoid sample, it was found that the neuron density of visual areas V1 and V2 scaled to brain mass to the $-1/3$ power (Sherwood et al. 2007), but that of area 9L did not correlate with brain mass (Sherwood et al. 2006, suppl.). The departures from brain size scaling trends documented here may reflect differences in the organization of neuronal connections related to specific functions, discussed below.

Recently, it has been suggested that cortical neuron numerical density scaling relationships actually vary between 2 mammalian orders: neuronal numerical density keeps pace with brain weight increase in primates, whereas in rodents, neuronal numerical density increases at a lower rate than brain

weight (Herculano-Houzel et al. 2007). This finding indicates that neuron density patterns vary between mammalian taxonomic groups, suggesting that brain weight is not the sole factor determining neuron density. However, the method used to draw this distinction in neuronal scaling pattern counted the entire cerebral cortex as a unit of analysis. Although it is hypothesized that the higher rate of neuronal density increase in primates allows for “greater computational power and cognitive abilities” (Herculano-Houzel et al. 2007), the study did not take into consideration organizational differences between rodent and primate cerebral cortices. For example, V1 has twice the neuron density of other cortical areas in primates, but not in other mammalian taxa (Rockel et al. 1980). Therefore, the large V1 volumes of the largest primate brains sampled by Herculano-Houzel et al. (cercopithecoids) might offset the trend of brain size-related decrease in neuronal density in the primate sample.

Studies which parcellate the cerebral cortex into functionally delimited regions may shed light on the more subtle causes of such scaling relationships. For example, the allometric scaling relationship between V1 and V2 neuron numerical density and brain weight found across anthropoids does not apply when limited to closely related species (Sherwood et al. 2007). In the current study, which included a taxonomically

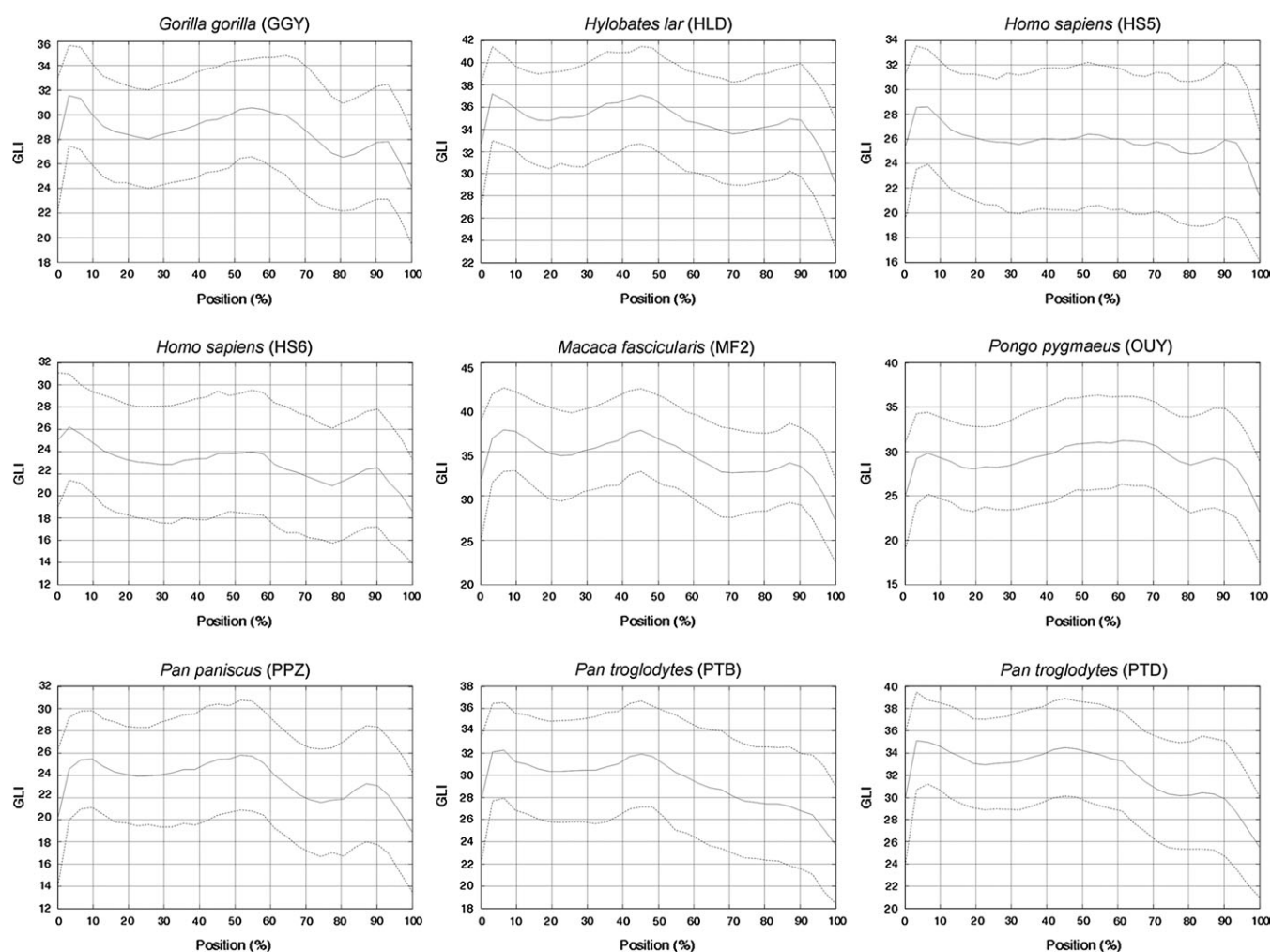


Figure 10. Mean V2 GLI profiles for each individual specimen (solid lines) with standard deviation indicated (dotted lines).

narrow sample, it was found that V1, V2, and VP mean GLI values did not correlate with any brain or body size variable. This, together with the GLI data discussed by Sherwood and Hof (2007), implies that within lower taxonomic groups, cell volume density is not constrained by brain or body size alone.

Although much attention has been paid to brain size's relationship to microanatomical variables, it has also been related to brain component volume (Finlay and Darlington 1995; Finlay et al. 2001). Alternatively, it has been found that brain component volumetric data scale to other components within a particular sensory system rather than to total brain size (Barton and Harvey 2000). This is explained by selection acting on particular functional systems within the brain, rather than the brain itself. The current study found that a histological variable, GLI, could be better explained by the size of functionally related visual system structures, than by brain size.

The sample here is taxonomically limited, and it is predicted that a relationship between GLI and brain size would exist for higher taxonomic groups, for example, across primates. However, interestingly, GLI values can be predicted by some visual system anatomical variables. This suggests that the total amount of input as well as details of interneuronal connections of a specific sensory modality, the visual system, may be the primary determinant of GLI values in visual areas. Thus, the more

fine-tuned differences between hominoid species in visual area cell volume density seem to be better explained by gross level size differences specific to the visual system.

In the clearest example in the current data set, an increase in V1 volume corresponds to a decrease in volume fraction of cell bodies (GLI), and, thus, an increase in volume fraction of neuropil. Cortical areas V2 and VP GLI values also scale negatively to V1 volume. This could be due to the fact that V1 serves as the primary source of visual inputs to extrastriate areas, and/or because these early extrastriate areas are, like V1, dependent on visual structures for input. In the case of V2, it is possible that this is a direct extrapolation of the scenario for V1, because V2 is the primary recipient of V1 inputs (Kuypers et al. 1965; Van Essen et al. 1986; Lund and Yoshioka 1991). Note that although it has been argued that, in macaques, VP (in contrast to V3d) is not a target of V1 inputs (Felleman and Van Essen 1991; Felleman et al. 1997), it has been pointed out that V2 projects symmetrically to V3d and VP (Gattass et al. 1997), therefore VP is certainly indirectly impacted by V1.

Interestingly, among visual cortex GLI values, only V1 GLI scales significantly against LGN volume. Perhaps this is because V1 is the first cortical recipient of most LGN projections in the geniculocalcarine pathway. However, although LGN is normally thought to serve as a "relay" for visual information that then

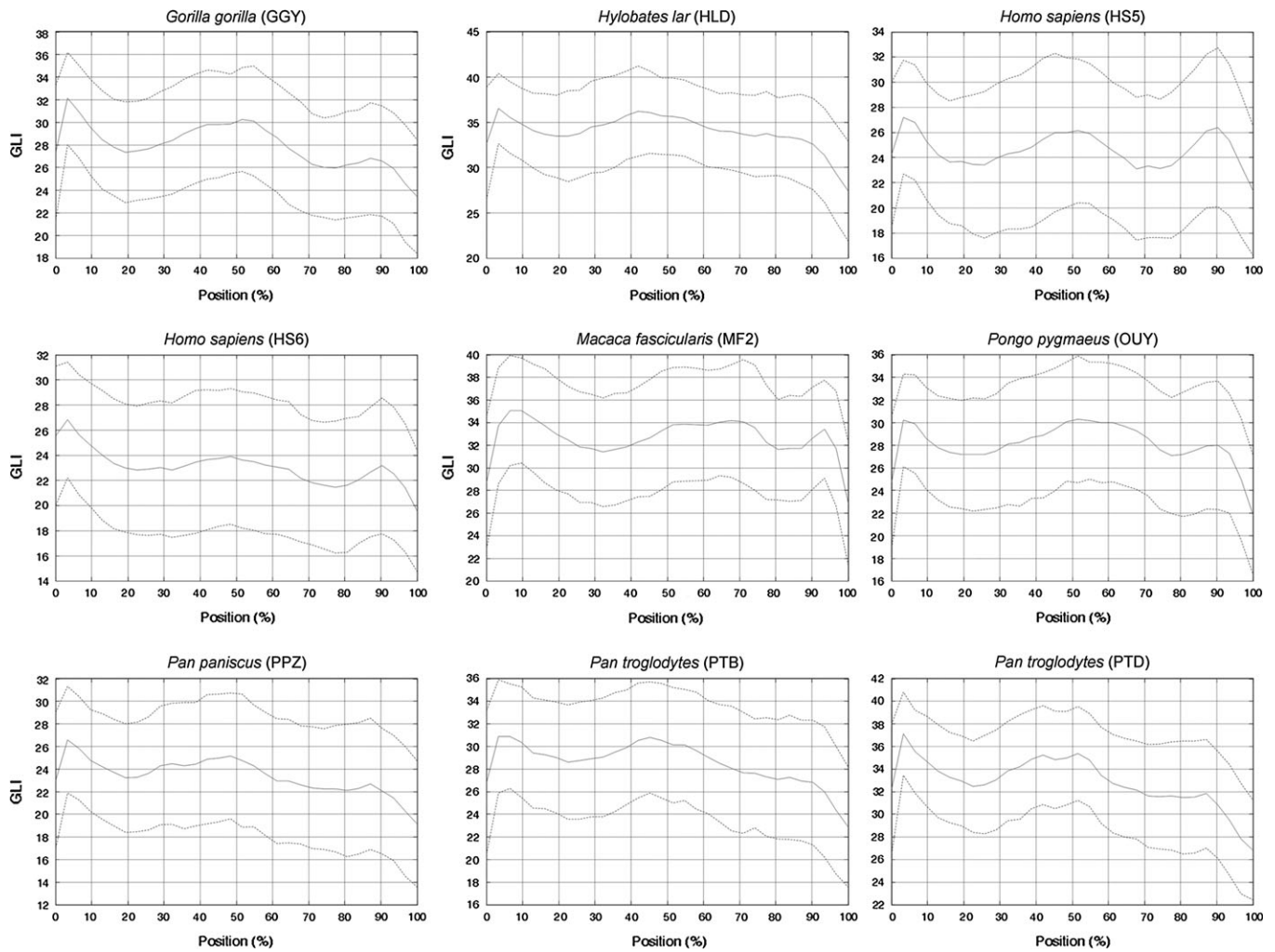


Figure 11. Mean VP GLI profiles of individual specimens (solid lines) with standard deviation indicated (dotted lines).

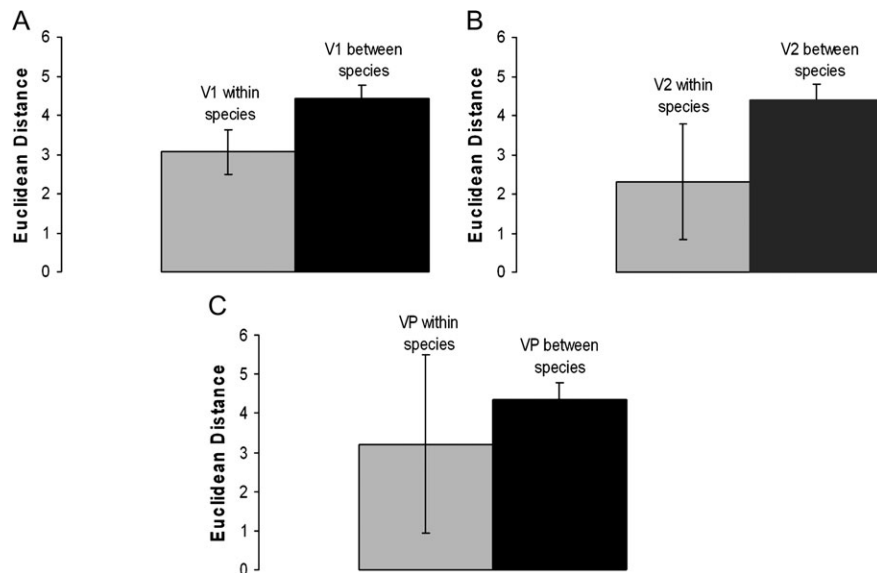


Figure 12. Bar graphs of Euclidean distances of GLI profile feature vectors among and within species (mean and 95% CI) for V1 (A), V2 (B), and VP (C).

goes to V1, in fact, V1 is not the sole cortical recipient of LGN inputs. A direct projection from LGN to V5 has been identified (Sincich et al. 2004), and additional direct connections between the LGN and extrastriate areas could exist.

There is a trend of decrease in GLI coincident with greater phylogenetic affinity to *Homo sapiens*, with the exception of *Pan troglodytes*. *Pan troglodytes* has also been found to have a high V1 neuronal density for a great ape (Sherwood et al. 2007). In contrast, *Pan paniscus*, the sister species of *Pan troglodytes*, has the lowest nonhuman GLI values for areas V1, V2, and VP, and is the only nonhuman to have GLI values within the human range. The intragenetic differences reported here between *Pan troglodytes* and *Pan paniscus* are interesting in combination with the observation that the 2 panin species also differ in regards to V1 volume (de Sousa et al., unpublished data). Similarly, intragenetic variation in the layout of visual cortical areas has been indicated for closely related species of macaques, which show variable morphology of the prelunate gyrus, in which V4 is positioned (Van der Gucht et al. 2006).

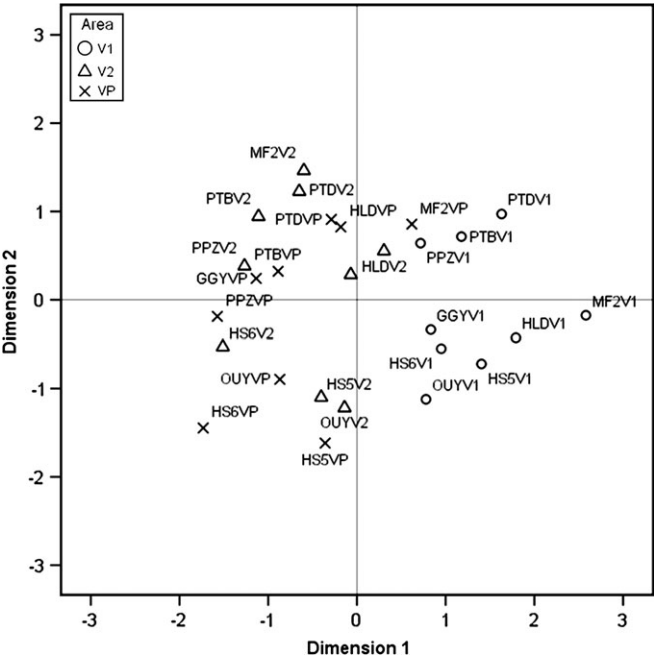


Figure 13. Euclidean distance plot of individual specimen GLI profiles based on a multidimensional scaling procedure. Minimum S-stress = 0.005.

Comparisons of Cortical Areas

Multivariate analyses demonstrated that GLI profile data can be used to compare cortical areas to each other in a multispecies sample. Area V1 was found to be the most distinct cortical area in terms of its GLI profile regardless of species, not surprising given its distinct laminar pattern. On the basis of just 3 feature vectors, it was possible to distinguish V1 from the extrastriate areas. Distinctions between extrastriate areas were found to be much more subtle. Visual area GLI profiles were also found to be more similar among members of the same species than among members of different species, as is the case with motor cortex (Sherwood et al. 2004). Finally, similarities between homologous extrastriate areas seemed to be blurred by species-specific and interindividual differences in cortical lamination. However, some aspects of GLI profiles showed potential for correctly sorting extrastriate areas into groups within the multispecific sample.

Overall, the analyses show that in terms of cytoarchitecture, V1 is more phylogenetically invariant than are the extrastriate areas. Because species-level extrastriate cortex cytoarchitectural differences might be based in the subtle differences of species-specific developmental trajectories, it is interesting to consider this observation in terms of the development of visual cortical areas. In enucleated fetal macaques, the region of cortex normally fated to become V1 develops cytoarchitectural characteristics of V2 (Rakic et al. 1991; Dehay et al. 1996), indicating a relationship between V1 cytoarchitecture and specific retinogeniculo inputs. The relative standardization of V1 cytoarchitectural pattern in the range of species studied here suggests that this relationship is conserved across species. The development of cytoarchitectonic differences between extrastriate areas, such as the increased columnarity of VP, or the denser layer IV in V2, is less clear. Note that V1 has firm cytoarchitectonic criteria which apply to the full extent of the area, and that it has clear borders with adjacent areas. Extrastriate area borders are so subtle that they were not defined in the present study, and although criteria were used to distinguish extrastriate areas from each other, their occurrence tended to be along a gradient, and was not uniform even within an individual.

Conclusions

The volume fraction of cell bodies, indicated by GLI, provides information about cortical neuronal organization across species. In this study, it was found that the laminar pattern of V1 is distinct from that of extrastriate areas in hominoids,

Table 3

Average euclidean distances between cortical areas V1, V2, and VP based on GLI profile feature vectors

A. Average distances between cortical areas, only comparisons per specimen ^a				Only within same specimen				
Area	V1	V2	VP	Mean	SD	No.	Conf.	Mean distance
V1	0.00			4.34	1.10	9	0.72	Between V1 and V2
V2	4.34	0.00		4.63	0.63	9	0.41	Between V1 and VP
VP	4.63	1.93	0.00	1.93	0.70	9	0.46	Between V2 and VP
B. Average distance between cortical areas, comparisons including all specimens ^b				Across all specimens				
Area	V1	V2	VP	Mean	SD	No.	Conf.	Mean distance
V1	3.51			4.93	1.07	81	0.23	Between V1 and V2
V2	4.93	3.65		5.15	1.04	81	0.23	Between V1 and VP
VP	5.15	3.55	3.75	3.55	1.37	81	0.30	Between V2 and VP

^aFor example, V2-VP average distance incorporates distance between HS5V2 and HS5VP, and distance between PPV2 and PPYVP.

^bFor example, V2-VP average distance incorporates distance between HS5V2 and HS5VP, and distance between GGYV2 and HLDVP.

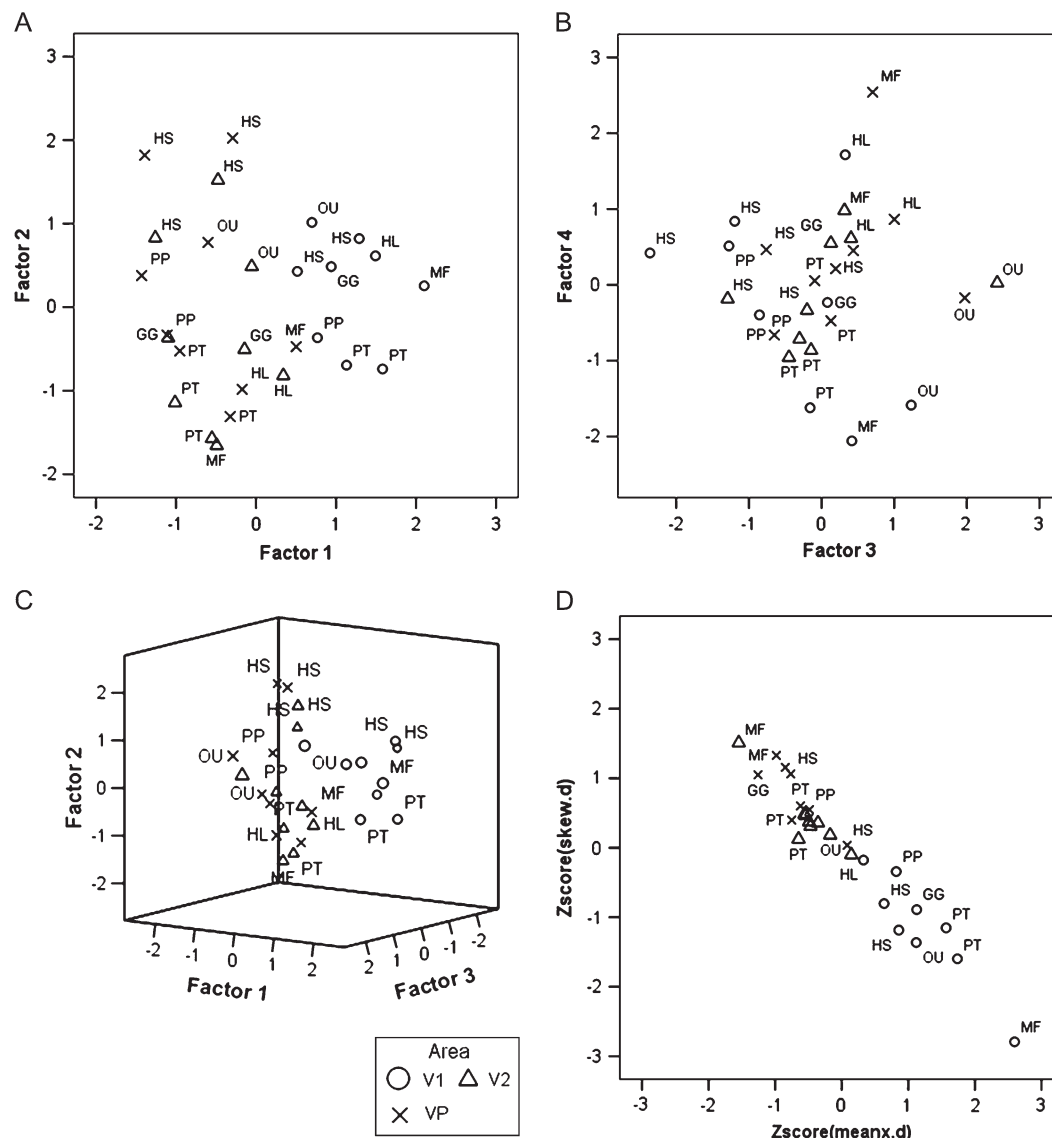


Figure 14. Principal components plots summarizing GLI individual profile data are shown (A,B and C). Also shown is a plot of the strongest positive (skew.d) and negative (meanx.d) loadings for Factor 1 (D). Abbreviations for species names are: HS - Homo sapiens, PT - Pan troglodytes, PP - Pan paniscus, GG - Gorilla gorilla, OU - Pongo pygmaeus, HL - Hylobates lar, MF - Macaca fascicularis.

Table 4

Component matrix of principal component analysis (PCA) of GLI profiles^a

	Component		
	1 (41.32%)	2 (26.36%)	3 (15.66%)
Zmeanx.o	0.66	-0.54	0.35
Zmeanx.o	0.69	0.39	0.5
Zsd.o	0.46	0.54	-0.55
Zskew.o	-0.56	-0.46	-0.59
Zkurt.o	-0.72	-0.44	0.41
Zmeanx.d	-0.67	0.64	-0.28
Zmeanx.d	0.81	0.26	-0.17
Zsd.d	0.61	-0.71	-0.19
Zskew.d	-0.76	-0.22	0.16
Zkurt.d	-0.36	0.69	0.46

^a3 components extracted using the principal component analysis extraction method.

and also in a macaque monkey. However, the organization of extrastriate areas V2 and VP is much less uniform, and interspecific and interindividual differences in laminar

organization overshadow distinctions between these cortical areas.

In addition, it was found that cortical visual area GLI values seem to be influenced by visual system organization and/or the total amount of visual input within this comparative sample. This has implications for the nature of brain organization. The results here suggest that total brain size is not sufficient for predicting differences in cortical area microanatomical organization among closely related species. Rather, the implication is that species-specific differences in aspects of histological organization, such as the density of neuronal connections, evolve as a part of functionally specific brain systems, and not as a function of overall brain size.

Supplementary Material

Supplementary material can be found at: <http://www.cercor.oxfordjournals.org/>

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Notes

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